29 October 2026
Editor-in-Chief
Human Reproduction

Dear Editor-in-Chief,

Please consider our manuscript, "Reference-country composition and age-standardized PCOS-attributed infertility contrasts: China and 18 G20 countries, 1990–2023", as an Original Article.

The scientific question is how much of the changing China-minus-international-reference difference in modeled PCOS-attributed infertility is allocated to changing country population shares within age, even when age-standardization weights remain fixed. This is relevant to readers using international burden estimates to monitor reproductive health: the reference population represented at each date is part of the question being answered.

In GBD 2023 estimates for females aged 15–49, the total China-minus-18-country difference changed by +151.54 per 100,000 between 1990 and 2023. Established symmetric accounting allocated +88.65 to reference-country composition, alongside +271.51 from Chinese rates and −208.62 from reference-country rates. Replacing reference shares with UN WPP 2024 gave a similar +86.76 composition component. Alternative benchmark definitions changed the sign of a near-zero secondary-infertility contrast in 2023. We report absolute components before ratios and distinguish demographic sensitivity from statistical uncertainty.

Liu and colleagues' 2024 *Human Reproduction* paper (doi:10.1093/humrep/dead241) described PCOS-attributed infertility burden and geographical trends. Our manuscript addresses the different question of how an explicitly pooled reference is constructed. Its contribution is a focused reproductive-epidemiology application and reporting implication, not a new decomposition algorithm or a claim that earlier national rate comparisons were erroneous. It does not estimate individual infertility risk, treatment benefit or causal effects.

The work is an exploratory secondary analysis of aggregate estimates. Confirmed jointly aligned GBD posterior draws were unavailable; uncertainty intervals for our custom contrasts and components are therefore not supplied. This limitation is explicit in the abstract and article. Code, aggregate results and a reproducibility archive are provided; licensed GBD originals and restricted source-cell extracts are excluded from the public repository. OpenAI-assisted coding and drafting are disclosed in the manuscript.

XX: Corresponding author to confirm final author approval, authorship order, exclusive consideration, related manuscripts/preprints, funding and disclosures, and institutional ethics-review applicability before submission. These declarations have not been assumed on behalf of the authors.

Yours sincerely,
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